

CYSE 301: Cybersecurity Techniques and Operations

Assignment 1: Traffic Tracing and Sniffing

Each student needs to login into the **CCIA virtual environment using the link** <https://cyber.lab.cova-cci.org/> to complete this assignment. You can find the VMs shortcuts for this lab on the Windows 10 desktop after logging in to the CCIA cyber lab.

Watch the recording posted under Module 1, page 1.3, to use Wireshark

AI Use Policy for This Assignment

You are encouraged — and in several places required — to use an AI assistant (e.g., ChatGPT, Gemini, Claude, or any other LLM of your choice) as a lab partner while completing this assignment. AI tasks are worth their own points.

Rules of engagement:

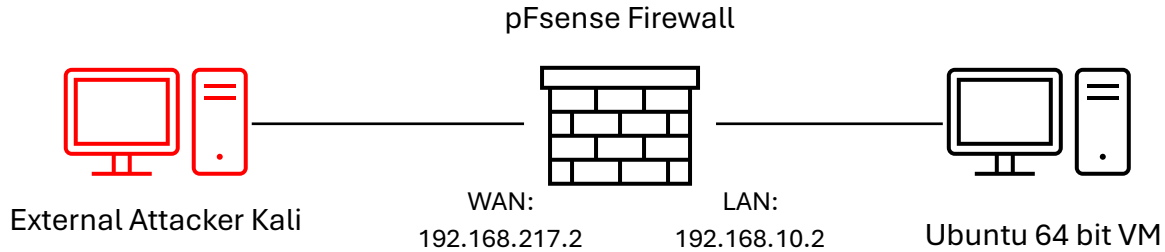
- AI may help you interpret packet captures, build/explain Wireshark filter syntax, explain protocol behavior, and draft written answers.
- AI may NOT perform the technical steps for you (running Wireshark, capturing traffic, extracting credentials, etc.). You must generate your own capture and screenshots on your own VMs provided by CCIA.
- You must independently verify every AI-generated claim against your own capture before submitting it. Screenshot the AI conversation for each question.
- Cite the AI tool and prompt style you used in the Tasks A and B reflection at the end of this assignment.

Submitting an unverified AI answer that contradicts/ supports your own packet capture will be treated as an incorrect answer, regardless of what the AI said.

Task A: Get started with Wireshark (35 points)

In this task, you will be using Wireshark **on External Kali** (NOT the one available in the CCIA Windows 10 Desktop) to monitor the traffic when External Kali and the Ubuntu VM are talking to each other.

This should look similar to the following diagram:



Please make sure to power on the pfsense VM at all times to keep the VMs connected.

You should turn on/ connect External Kali and also connect /turn on the Ubuntu VM.

TODO:

Follow each step in sequence:

1. Open Wireshark on External / Attacker Kali and listen on interface “eth0”.
2. keep Wireshark **in External / attacker Kali** running in the background while performing the following subsequent tasks 3 and 4.
3. Open a new terminal in Attacker Kali, then ping the Ubuntu VM for 5 – 10 seconds. You can stop ping by hitting CTRL + C key.
4. Open a new web browser in Attacker Kali (even if no webpage will be displayed) and keep it for a couple of seconds.
5. **Stop capturing** (by clicking the **red button** on the tool bar).
6. **Save the pcap file in root folder in the External/Attacker kali.**

Now, answer the following questions based on the packets captured in Wireshark.

(**Screenshot required for the Wireshark filter results** that proves the answers to each question you write here — *label it with the exact step/question number. Eg, Task-A-Q1*)

Q1. (5 pts) How many packets are captured in total? How many packets are displayed?

Q2. (5 pts) Apply “ICMP” as a display filter in Wireshark. Then repeat the previous question (Q1).

Q3. (5 pts) Select an Echo (reply) message from the list. What are the source and destination IPs of this packet? What are the sequence number and the size of the data? What is the response time?

Q4. (5 pts) Apply “DNS” as a display filter in Wireshark. How many packets are displayed?

Q5. (5 pts) Find a DNS query packet. What is the domain name this host is trying to resolve? What is the source IP and port number, destination IP and port number?

Q6. (5 pts) Find the **corresponding** DNS response to the query you selected at the previous step (Q-5), and what is the source IP and port number, destination IP and port number? What is the message replied from the DNS server?

(Using AI) Q7. (5 points) In your local computer,

- Use any AI tool (GPT, Gemini, Claude, or any other LLM).
- Prompt your AI tool / assistant to explain, in its own words, the difference between the ICMP Echo Request and Echo Reply message types, and why DNS typically uses UDP port 53. Compare the AI's explanation to what you actually observed in your capture — does it match? Note any discrepancy.

-----**End of Task-A**-----

Task B: Sniff LAN traffic (65 Points)

In this task, you will be acting as an **Analyst** who sniffs the regular communications between peers/sources (External Attacker Kali and Ubuntu) by using Wireshark on **Internal Kali VM** (Destination)

I would recommend you keep Wireshark running on Internal Kali all the time.

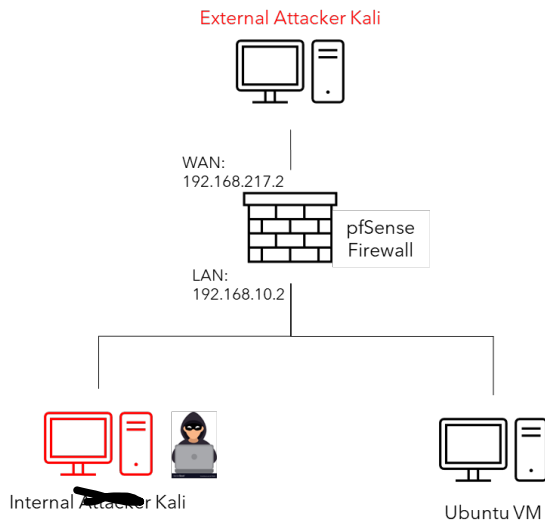


Figure 2 Required VMs for this assignment

IMPORTANT UPDATE: Since Port Mirroring is enabled to "Source" in Attacker Kali and Ubuntu and "Destination" in Internal Kali, you don't need to do any settings.

Screenshot required — *label all the screenshots for the following tasks/steps with the exact step/question number. Eg, Task-B-1a*

TODO:

1. Sniff ICMP traffic (5+ 5 + 5 + 5 + 10 = 30 points)

a) (5 points) Please connect to the following VMs in Hyper-V Manager:

- Attacker/External Kali,
- Internal kali,
- pfsense, and

- Ubuntu.
- b) **(5 points)** Launch and run Wireshark in Internal Kali.
 - c) **(5 points)** Open two terminals on External Kali VM. Use one to ping Ubuntu VM, and use the other to ping Internal Kali.
 - d) **(5 points)** Apply proper display or capture filter in Wireshark on **Internal Kali VM** to show active ICMP traffic.
 - e) **(10 points)** Apply a proper display or capture filter on the internal Kali VM that **ONLY** displays the ICMP request that originated from the external Kali VM and goes to Ubuntu 64-bit VM.

2. Sniff FTP traffic (5 + 10 +10 = 25 pts points)

- a. **Ubuntu VM** is also serving as an FTP server inside the LAN network. Now, you need to use External Kali to access this FTP server by using the command: **ftp [ip_addr of ubuntu VM]**. The username for the FTP server is **student**, and the password is **password**. You can follow the steps below to access the FTP server.

```
(root@kali)-[~]
# ftp 10.10.10.10
Connected to 10.10.10.10.
220 (vsFTPd 3.0.5)
Name (10.10.10.10:root): student
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> exit
221 Goodbye.
```

- b. **Unfortunately**, Internal Kali, the attacker, is also sniffing into communication. Therefore, all of your communication is exposed to the attacker. Now, you need to find out the **password** used by External Kali to access the FTP server from the intercepted traffic on Internal Kali. You need to take a screenshot and explain how you found the password.
 - c. After you successfully find the username & password from the FTP traffic, repeat the previous step (2.a), and use your **MIDAS ID** as the username and **UIN** as the password to re-access the FTP server from External Kali. Although External Kali may not access the FTP server, you need to intercept the packets containing these “secrets” from the attacker VM, which is **Internal Kali**.
- ## 3. Root Cause and Mitigation (Using an AI tool) (10 points)

Paste the output with FTP command/response you extracted for task-B, Q2-b from Wireshark running in Internal Kali, into your AI assistant (as text, **not** for task-b-Q2c which is your live credential) and ask it to explain why FTP exposes credentials in plaintext and what protocol(s) or configuration changes would prevent this exact sniffing attack (e.g., FTPS, SFTP, VPN, switched-network segmentation, encrypted tunneling). **Write a 4–6 sentence mitigation recommendation** in your own words, citing the AI's suggestions but validating them against what you learned about port mirroring in task-B.

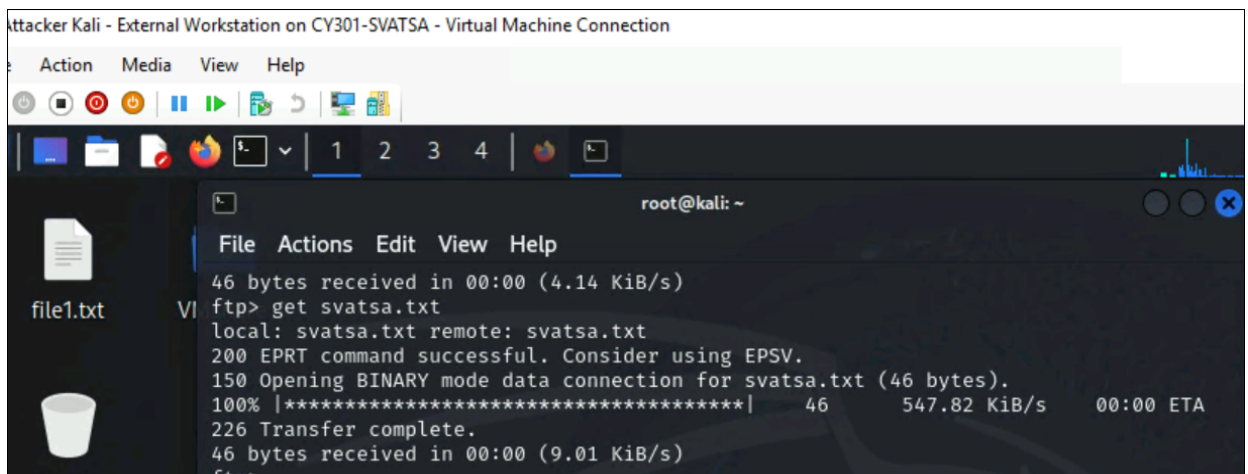
-----**End of Task-B**-----

Extra credit: Steal files with Wireshark (15 points)

Login to Ubuntu VM, and create a file in your home directory named “YOUR_MIDAS.txt”. Put the **current timestamp** and **your name** in the file. You can use the following command in the example below to do the job.

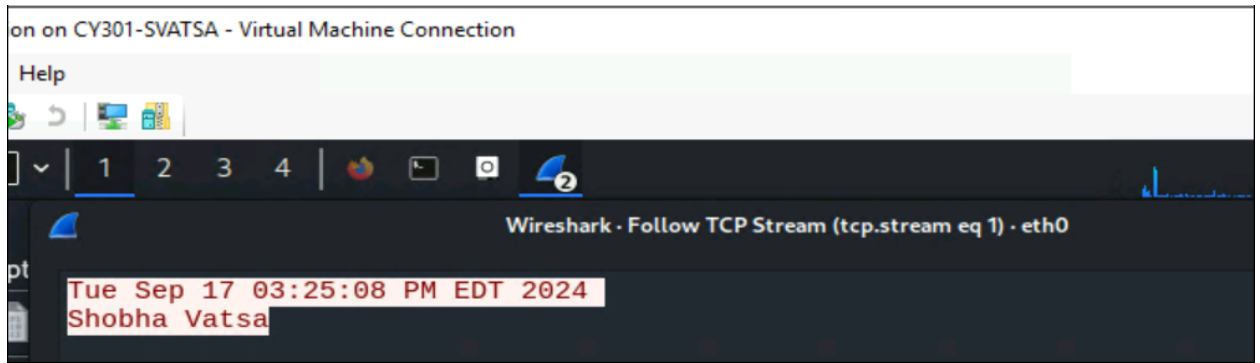
```
student@ubuntu:~$ echo -e "$(date) \nShobha Vatsa" > svatsa.txt
student@ubuntu:~$ ls
Desktop      examples.desktop  newDir1      Public  svatsa.txt  Videos
Documents    forVatsa.txt      passwd_Vatsa snap     Templates  VMshare
Downloads    Music             Pictures      svatsa   Vatsa
student@ubuntu:~$ cat svatsa.txt
Tue Sep 17 03:25:08 PM EDT 2024
Shobha Vatsa
student@ubuntu:~$
```

Once you have the file ready in Ubuntu, switch back to **External Kali**. Get the file you just created remotely using the FTP protocol. Below is an example.



As an attacker, you need to complete the following tasks in Internal Kali:

1. Apply a proper display filter to display the **FTP-DATA** packets between External Kali and the Ubuntu VM.
 2. Follow the TCP stream of the FTP-DATA packet and view the content of the file just transferred.
 3. Export (Save) the transferred file as a text file in Internal Kali and view the content.
- Below is an example.



Deliverables:

- Screenshot for each task/ question asked with relevant task and label/step numbers
- **Create a technical report detailing the tasks you performed, the prompts you used, and the results you received from the LLM. Make sure the report contains a link to your conversation with the LLM.**
- **Note:** ChatGPT allows you to publicly share a conversation. Learn more about the sharing options here: <https://help.openai.com/en/articles/7925741-chatgpt-shared-links>**Links to an external site.**. If you use a different LLM, figure out how to share the conversation.